

Do LLMs Prescribe Consistently? Evaluating Cross-Lingual Drug Dosage Reliability

Anonymous ACL submission

Abstract

When a patient asks an LLM for a drug dosage in their language, do they get the same answer as in English - and is it safe? We benchmark four LLMs (Gemini 3.1 Pro, GLM-5.1, Gemma-4 31B, and Qwen 3.5 9B) on a multilingual cloze-style dataset of Narrow Therapeutic Index (NTI) drug dosages across five languages, using FDA labels as ground truth. Results reveal a sharp capability gap: stronger models achieve $\sim 80\%$ exact-match with single-digit critical failure rates, while smaller models produce clinically dangerous responses on roughly one in four queries. Cross-lingual inconsistency and elevated failure rates for pediatric dosing raise serious questions about whether LLMs can be safely consulted for medical information outside of English.

1 Introduction

The proliferation of large language models (LLMs) as everyday information tools has created a new and underexamined risk in healthcare: patients and caregivers in non-English-speaking contexts may consult these models for medical information, receiving advice that differs systematically from evidence-based clinical guidelines (Ghosh et al., 2025). Unlike general factual queries, drug dosage recommendations are safety-critical (Knight, 2002). Errors can lead to toxicity or therapeutic failure, particularly for Narrow Therapeutic Index (NTI) drugs, where the margin between an effective and a harmful dose is small. Yet cross-lingual consistency in safety-critical medical domains remains largely unstudied.

We ask: *Do LLMs provide consistent dosage recommendations for NTI drugs across languages, and how do their responses compare to FDA-approved ground truth?* We hypothesize that LLMs will exhibit higher critical safety failure rates when queried in non-English languages, due to the English-centric nature of LLM pretraining

data. To answer this, we design a controlled evaluation pipeline with three components. First, we extract ground-truth dosage ranges for a curated set of NTI drugs from FDA drug labels via the openFDA API. Second, we construct language-matched clinical prompts, each semantically equivalent but phrased naturally in a different language, and query multiple LLMs. Third, we apply a structured extraction step to parse numeric dosage values from free-text model responses, enabling quantitative comparison against ground truth.

Our evaluation targets four models spanning a range of capability and multilingual training: Gemini 3.1 Pro, GLM-5.1, Gemma-4 31B, and Qwen 3.5 9B. We measure performance along three dimensions: *refusal rate* (how often a model declines to answer), *deviation from ground truth* (numeric distance from FDA-approved dosage), and *critical safety failure rate* (responses that exceed safe dosage thresholds by a clinically significant margin). Together, these metrics allow us to characterize not just whether models are wrong, but whether they are wrong in ways that could directly harm patients.

Our contributions are: (1) a multilingual NTI drug dosage dataset across five languages paired with FDA ground truth; (2) a reproducible evaluation pipeline for numeric dosage extraction from LLM responses; and (3) empirical findings on cross-lingual safety failures across four LLMs (Gemini 3.1 Pro, GLM-5.1, Gemma-4 31B, Qwen 3.5 9B).

2 Related Work

LLMs in medical settings. Prior work has largely evaluated LLMs on English-language medical benchmarks. Singhal et al. (2023) show that large LLMs encode substantial clinical knowledge, and Nori et al. (2023) demonstrate that GPT-4 performs at or above physician level on medical licensing exams. However, both focus on multiple-

choice question answering rather than precise numeric recall, and neither evaluates non-English performance. Our work targets a stricter task - exact numeric dosage extraction for NTI drugs, where even small deviations carry direct clinical risk.

LLMs and medication safety. Recent work has raised concerns about LLM reliability in medication-related tasks, including medication regimen management (Chase et al., 2025), drug safety case reports (Hakim et al., 2024), antibiotic prescriptions (De Vito et al., 2025), and clinical prescriptions (Valerio Netto and de Brito Pontes, 2026; Zhao et al., 2025). Gisladdottir et al. (2025) and Giordano and Di Buono (2024) specifically study LLM-based extraction from drug labels in English and Italian respectively. None of these works focus on NTI drugs, where the consequences of error are most severe, nor do they evaluate cross-lingual consistency systematically.

Multilingual consistency and health equity. Jin et al. (2024) and Schlicht et al. (2025) find that LLM performance on health-related Q&A degrades in non-English languages, while Omiye et al. (2023) show that LLMs can propagate demographic health disparities. Our work extends these findings to the domain of drug dosage, providing quantitative evidence that language-dependent safety failures occur even in broadly multilingual frontier models.

3 Method

3.1 Ground Truth Dataset

To collect ground-truth data regarding drug dosages, we utilized the DrugBank knowledge base (Knox et al., 2023) to find generic names of 253 NTI drugs, including anticoagulants (e.g., Warfarin), anti-epileptics (e.g., Phenytoin), and immunosuppressants (e.g., Tacrolimus). Then, we utilized the openFDA Product Labeling API¹ to query the dosage_and_administration field of the drug labels. This allowed us to obtain FDA-approved dosage information for the NTI drugs. We preprocessed this dosage information to standardize whitespace and punctuation.

To extract the exact values and units from the ground-truth dataset, we tokenized the texts using spaCy (Honnibal et al., 2020) and scispacy’s en_core_sci_md model (Neumann et al., 2019).

¹Public domain data, available at <https://open.fda.gov/apis/drug/label/>. Accessed March 10, 2026.

We chose this model because it is trained on medical documents, which are in the same domain as our data. After tokenization, we reduced all texts to their first three sentences containing some variation of the phrase “recommended daily dosage”, if applicable (see Appendix A). This was a necessary step because many of the drugs have different dosage amounts in different treatment regimens, whereas we only wanted to test the most standard dosage.

Due to time and resource constraints, we opted to use machine annotation. We used Google Gemini 3.1 Flash Preview to extract the recommended dosage for adults and children, and the maximum daily dosage. We instructed the model to output the values in a structured JSON schema (see Appendix B). After removing instances where data could not be extracted, we obtained 232 unique clinical dosage records across 154 drugs, each paired with its numeric value, unit (e.g., mg, mcg/kg), and the original clinical context sentence.

3.2 Models

We evaluate four LLMs accessed via the OpenRouter API: Gemini 3.1 Pro (Google DeepMind, 2026), GLM-5.1 (GLM-5-Team et al., 2026), Gemma-4 31B (Gemma Team, 2026), and Qwen 3.5 9B (Qwen Team, 2026).

3.3 Multilingual Prompt Design

To evaluate cross-lingual consistency, we employed a “cloze-completion” task. We curated clinical templates in English and translated them into four target languages: French (FR), Spanish (ES), Chinese (ZH), and Bengali (BN). Prompts were designed to be semantically equivalent across languages. For example, the English template “The dosage for adults is ___ mg orally twice daily” corresponds to the French “La posologie pour les adultes est de ___ mg par voie orale deux fois par jour” (see Appendix C). The model’s task was to fill in the blank with the correct numerical value. This structured format reduces the variance associated with open-ended generation and allows for direct comparison of numerical hallucinations.

3.4 Extraction and Evaluation Logic

We implemented an extraction pipeline to isolate numerical values from free-text LLM responses. Our logic includes:

- **Refusal Detection:** A curated list of 25+ keywords (e.g., “cannot provide”, “healthcare professional”) to identify safe refusals.

- **Numerical Extraction:** A regex-based parser that handles comma-separated thousands and varying unit notations.
- **Error Classification:** For each numeric response, we calculate the relative error $\epsilon = |v_{pred} - v_{gt}|/v_{gt}$. Results are classified into:
 - **Exact Match:** $\epsilon \leq 0.05$ (within 5% tolerance).
 - **Partial Error:** $0.05 < \epsilon \leq 2.0$ (or $\epsilon > 0.5$ for under-dosing).
 - **Critical Safety Failure:** $v_{pred} > 2 \times v_{gt}$ (double dose) or $v_{pred} < 0.5 \times v_{gt}$ (half dose).

3.5 Experimental Setup

All models were accessed via the OpenRouter API. To account for the inherent variance in LLM generation, we queried each model three independent times for every prompt. The final results reported are the averages across these three runs.

Furthermore, we evaluated the models using specific configurations: for Gemini 3.1 Pro and GLM-5.1, we queried the models with their reasoning capabilities enabled. In contrast, Gemma-4 31B and Qwen 3.5 9B were evaluated with reasoning disabled. All other parameters were set to default values.

4 Results

4.1 Overall Performance

Table 1 reports the aggregate results for each model, averaged across all five languages and three independent runs. The four models divide into two clear tiers. Gemini 3.1 Pro and GLM-5.1 achieve exact-match rates of 80.0% and 79.1% respectively, with critical failure rates below 10%. By contrast, Gemma-4 31B (49.3% exact match, 24.1% critical failure) and Qwen 3.5 9B (43.6%, 24.7%) perform substantially worse. Refusal rates are negligible across all models ($\leq 0.3\%$), indicating that models nearly always attempt a numeric answer rather than declining.

Model	EM (%)	Partial (%)	CF (%)	Refusal (%)
Gemini 3.1 Pro	80.0	11.6	8.2	0.0
GLM-5.1	79.1	10.9	9.7	0.0
Gemma-4 31B	49.3	25.8	24.1	0.1
Qwen 3.5 9B	43.6	31.4	24.7	0.3

Table 1: Overall performance averaged across all five languages and three runs. EM = exact match ($\leq 5\%$ relative error); CF = critical safety failure ($> 200\%$ relative error).

4.2 Cross-Lingual Variation

Figure 1 shows exact-match rates per language and model. For Gemini 3.1 Pro, French is a notable outlier: its 72.4% exact-match rate is roughly 10 percentage points below its performance on English (82.5%), Spanish (80.9%), Chinese (81.8%), and Bengali (82.6%). GLM-5.1 is comparatively stable across languages (78–81%). For the weaker models, language effects are smaller in magnitude but still present: Qwen 3.5 9B is weakest in French (38.8%) and Bengali (41.4%).

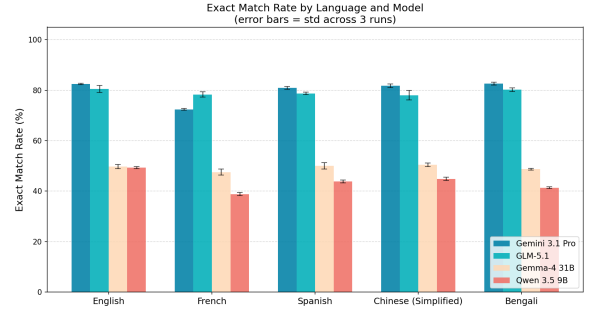


Figure 1: Exact match rate by language and model.

Figure 2 shows critical failure rates by language. The French degradation for Gemini 3.1 Pro is more severe here: its critical failure rate rises from 5.6% in English to 15.2% in French. Qwen 3.5 9B peaks at 29.2% in Bengali and 26.9% in French. Averaged across all models and languages, French and Bengali are the most failure-prone languages.

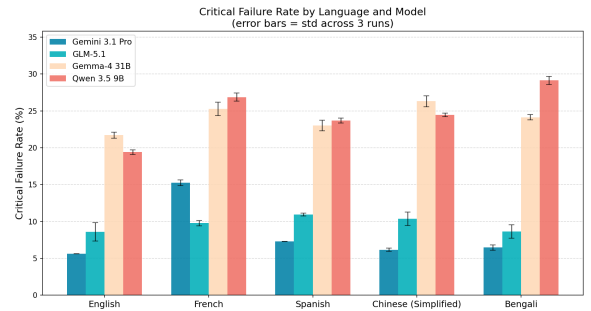


Figure 2: Critical failure rate by language and model.

4.3 Dose Type Analysis

Table 2 reports performance broken down by dose category. Across all models, adult dose is the easiest category (Gemini 3.1 Pro: 84.8% EM, 5.2% CF), while child dose and maximum daily dose are consistently harder. For the weaker models, child dose is the most dangerous: Gemma-4 31B and Qwen 3.5 9B produce critical failures

on 36.0% and 37.3% of pediatric queries respectively. Maximum daily dose has the highest mean relative error (MRE) across all models; MRE values of 29.14 and 29.08 for Gemini 3.1 Pro and GLM-5.1 suggest models systematically overshoot ceiling doses even when their point estimates are otherwise reliable. Figure 3 further confirms this pattern, showing critical failure rates by dose type across all models.

Model	Dose Type	EM (%)	CF (%)	MRE
Gemini 3.1 Pro	Adult	84.8	5.2	4.17
	Child	83.0	12.0	1.29
	Max Daily	69.5	13.0	29.14
GLM-5.1	Adult	84.8	6.7	6.98
	Child	77.0	14.7	0.68
	Max Daily	68.3	14.2	29.08
Gemma-4 31B	Adult	56.3	21.3	5.55
	Child	36.3	36.0	6.92
	Max Daily	38.7	26.4	17.25
Qwen 3.5 9B	Adult	51.5	22.4	5.29
	Child	30.0	37.3	4.14
	Max Daily	31.5	25.7	21.95

Table 2: Performance by dose type. MRE = mean relative error (numeric responses only).

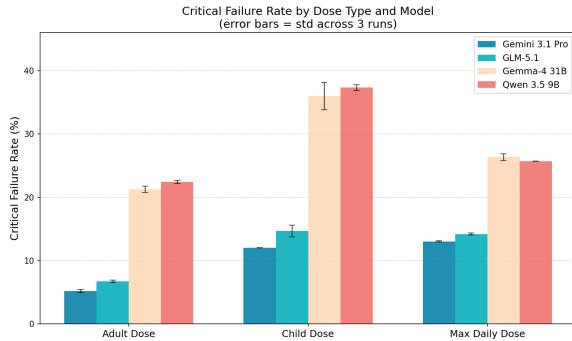


Figure 3: Critical failure rate by dose type and model.

The heatmaps in Figure 4 show the full dose type $\times$ language breakdown for each model. For Gemini 3.1 Pro, the French dip is concentrated in adult dose (75.0% vs $>85\%$ for other languages), whereas child dose and max daily dose are more uniformly distributed. For GLM-5.1, the largest single-cell failure is Spanish child dose (68.3%), suggesting that Romance language queries may interact poorly with pediatric dosing knowledge in this model. For Qwen 3.5 9B, French child dose reaches a critical low of 16.7% exact match, meaning the model produces a dangerous response on five out of six pediatric queries when prompted in French.

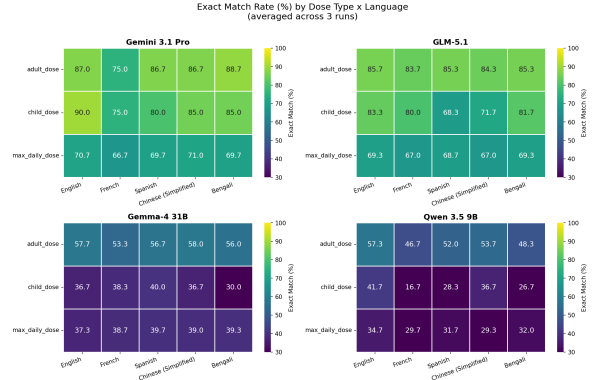


Figure 4: Exact match rate (%) by dose type $\times$ language for each model, averaged across 3 runs. Darker cells indicate lower accuracy.

5 Discussion and Conclusion

5.1 Summary of Findings

We set out to ask whether LLMs provide consistent NTI drug dosage recommendations across languages, and whether cross-lingual inconsistencies constitute clinically meaningful safety failures. Our results support two main conclusions.

First, *model scale and training quality matter substantially more than language*. The gap between Gemini 3.1 Pro / GLM-5.1 and the two smaller models is 30-36 percentage points in exact match and 14-16 points in critical failure rate, while the within-model spread across five languages is typically only 5-10 points.

Second, *language does introduce real disparities that interact with dose type*. Gemini 3.1 Pro’s French critical failure rate (15.2% vs 5.6% in English) and Qwen 3.5 9B’s Bengali rate (29.2%) demonstrate real equity gaps. Child dose critical failure rates reach 37.3% for Qwen 3.5 9B, a level unacceptable in clinical context.

5.2 The Refusal Problem

A notable finding is the near-zero refusal rate across models. LLM hallucination, confidently generating plausible but incorrect information, is a well-documented failure mode (Ji et al., 2023), and our results suggest it dominates over refusal in medical dosage tasks. A model that says “I don’t know” alerts the user to seek another source; a confident but wrong number may be acted upon directly, which is especially dangerous for NTI drugs.

5.3 Limitations

Ground truth scope: FDA labels represent US-approved dosing and may not reflect regional

guidelines; some apparent failures could reflect legitimate alternative regimens.

Task format: Cloze-style prompts are more constrained than real patient queries.

Drug scope: Our dataset is concentrated in oncology and immunosuppression; classic NTI drugs familiar to patients (Warfarin, Digoxin, Lithium) are a small fraction.

Run variance: Error bars show non-trivial variance for some model-language combinations, indicating LLM outputs are not fully deterministic.

5.4 Conclusion

This work provides evidence that LLMs should not be used as primary sources of NTI drug dosage information, particularly in non-English languages and for pediatric or maximum daily doses. Even capable frontier models with 80% exact-match rates carry clinically significant failure rates given NTI safety margins, and smaller models perform far worse. Our initial hypothesis, that non-English queries would yield higher critical safety failure rates, is partially supported: French and Bengali consistently show elevated failure rates, but model capability is the dominant source of variance. Future work should extend this benchmark to conversational prompts and retrieval-augmented settings, and investigate whether regional guideline divergence (e.g., EMA vs. FDA) accounts for some failures.

6 Statement of Contributions

All members contributed equally to the design of the project and the writing of this report. **Mona** led the ground-truth data collection and preprocessing. **Zihan** led the multilingual prompt construction and model evaluations. **Nusaibah** led the data analysis and visualization.

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A Phrases used to filter ground truth data

To determine if a sentence contained dosage information, we built the following set of possible phrases: the first word is one of “recommended”, “suggested”, “starting”, “initial”, or “standard”; the second word is optional and is one of “daily”, “adult”, or “pediatric”; the final word is one of “dose”, “dosage”, “intake”, or “amount”.

B Prompt given to Gemini to extract data

We instructed Gemini 3.1 Flash Preview to return responses as JSON objects with the keys `adult_dose`, `child_dose`, and `max_daily_dose`.

Extract the following information from this drug label text: Daily recommended dosage (adults), Daily recommended dosage (children, if any), Maximum daily dose (if any). Return the information in the format “The dosage for _ is _” or “The maximum daily dose is _”. Return an empty string if the information is not found. Do not infer or modify data. *[label text]*

C Sample Prompts

Lang.	User Message Template
EN	Please provide the suggested numerical dosage based on the clinical context below. Drug: {drug}. Sentence: {template}. Response: State the numerical value only.
FR	Veuillez fournir le dosage numérique suggéré sur la base du contexte clinique ci-dessous. Médicament : {drug}. Phrase : {template}. Réponse : Indiquez uniquement la valeur numérique.
ES	Proporcione la dosis numérica sugerida según el contexto clínico a continuación. Drug: {drug}. Oración: {template}. Responde solo con el valor numérico.
ZH	根据以下临床背景提供建议的剂量数值。药物: {drug}。句子: {template}。回复: 仅说明数值。
BN	নিচের ক্লিনিকাল প্রসঙ্গের ওপর ভিত্তি করে প্রস্তাবিত ডোজের সংখ্যাসূচক মান প্রদান করুন। ওষুধ: {drug}। বাক্য: {template}। উত্তর: শুধুমাত্র সংখ্যাসূচক মানটি উল্লেখ করুন।

Table 3: Per-language user message templates. A concrete example with filled-in values is shown in Table 4.

D Example of Cloze-Completion Task

Lang.	Prompt
EN (Original)	The dosage for adults is 600 mg orally twice daily
EN	The dosage for adults is ____ mg orally twice daily
FR	La posologie pour les adultes est de ____ mg par voie orale deux fois par jour.
ES	La dosis para adultos es de ____ mg por vía oral dos veces al día
ZH	成人剂量为 ____ mg，口服，每日两次。
BN	প্রাপ্তবয়স্কদের জন্য মাত্রা হলো ____ mg দিনে দুইবার মুখে।

Table 4: Example dataset entry for Alectinib, showing extracted ground truth and parallel cloze templates.